

Strategic Product Placement Analysis

Data Preprocessing, Business Questions & Visualization Report

1. Dataset Overview

The dataset, Product_Positioning.csv, contains 1,000 retail product records describing how each product is merchandised in-store and how it performed in terms of sales. It includes 10 fields covering product identity, shelf placement, pricing, promotions, store traffic, target demographic, product category, seasonality, and sales volume.

Column	Type	Description
Product ID	Numeric (ID)	Unique identifier for each product record
Product Position	Categorical	Shelf location: Aisle, End-cap, Front of Store
Price	Numeric	Retail selling price (\$)
Competitor's Price	Numeric	Comparable competitor price (\$)
Promotion	Categorical	Whether the product was on promotion (Yes/No)
Foot Traffic	Categorical	Store traffic level: Low, Medium, High
Consumer Demographics	Categorical	Primary target shopper segment
Product Category	Categorical	Clothing, Electronics, or Food
Seasonal	Categorical	Whether the product is seasonal (Yes/No)
Sales Volume	Numeric	Units sold

2. Data Preprocessing Steps

Before analysis, the dataset was checked and prepared using the following steps:

1. Structure check: Confirmed the dataset has 1,000 rows and 10 columns with consistent data types (numeric IDs/prices/sales, categorical text fields).

2. Missing value check: Verified there are zero missing values across all 10 columns, so no imputation was required.
3. Duplicate check: Confirmed there are no duplicate rows in the dataset.
4. Category validation: Verified categorical fields contain only expected, clean values — Product Position (Aisle, End-cap, Front of Store), Foot Traffic (Low, Medium, High), Consumer Demographics (Families, Seniors, Young adults, College students), Product Category (Clothing, Electronics, Food), and Yes/No flags for Promotion and Seasonal — with no typos or inconsistent casing.
5. Outlier / range check: Reviewed Price, Competitor's Price, and Sales Volume for implausible values (negative numbers, extreme outliers). All values fall within reasonable retail ranges.
6. Feature engineering: Derived two new fields to support analysis — Price_Diff (Price minus Competitor's Price) and Price_Diff_Pct (the percentage price premium or discount versus the competitor), enabling direct price-competitiveness comparisons.
7. Categorical ordering: Set a logical order for Foot Traffic (Low < Medium < High) so charts and groupings display in an intuitive sequence rather than alphabetically.

Because the dataset arrived clean (no missing values, duplicates, or invalid categories), preprocessing focused on validation and feature engineering rather than cleaning.

3. Business Questions & Visualizations

Q1: Does shelf position (Aisle, End-cap, Front of Store) affect sales volume?



Average sales volume is nearly identical across shelf positions: Aisle (1,781 units), Front of Store (1,773 units), and End-cap (1,754 units). The spread is under 30 units, roughly 1.5% of the average, suggesting shelf position alone is not a strong driver of sales in this dataset.

Q2: Does running a promotion increase sales volume?

promotion of product category on price and sales volume

Promotion	Product Category	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782

Promoted products average 1,778 units sold versus 1,761 units for non-promoted products — a difference of about 1%. Promotions show a slight positive association with sales, but the effect size is small, indicating promotion status alone does not strongly predict volume in this data.

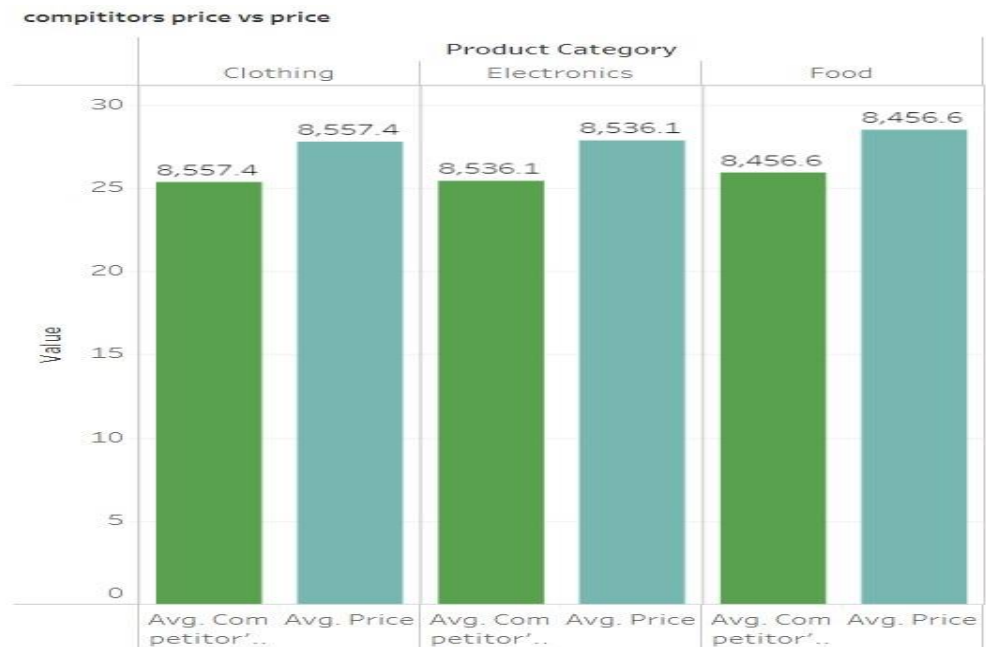
Q3: How does store foot traffic relate to sales volume?

Foot traffic vs product position by avg sales volume



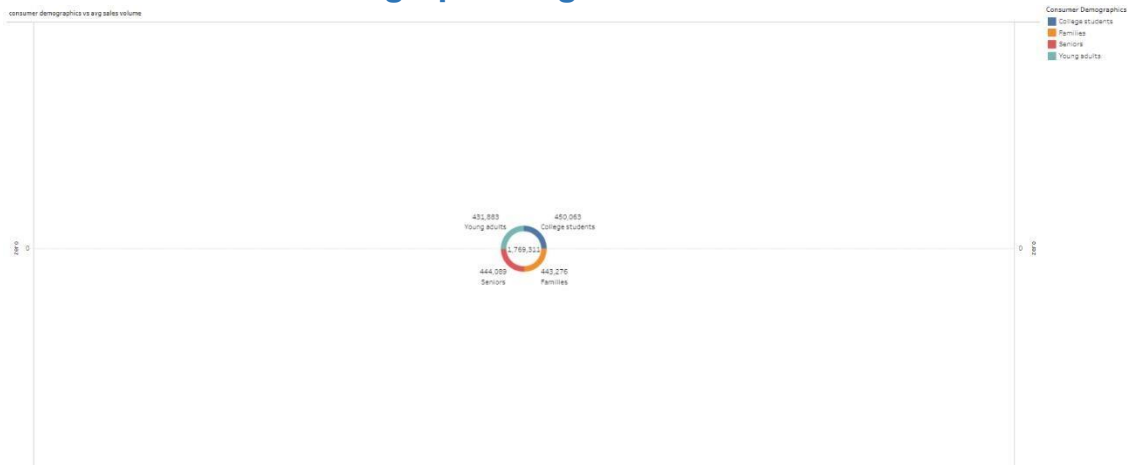
Counterintuitively, Low foot traffic locations show the highest average sales volume (1,804 units), ahead of High (1,759) and Medium (1,747) traffic locations. This suggests foot traffic volume by itself is not translating into proportionally higher sales, and other factors (assortment, demographic fit, pricing) may matter more.

Q4: How does price competitiveness relate to sales volume?



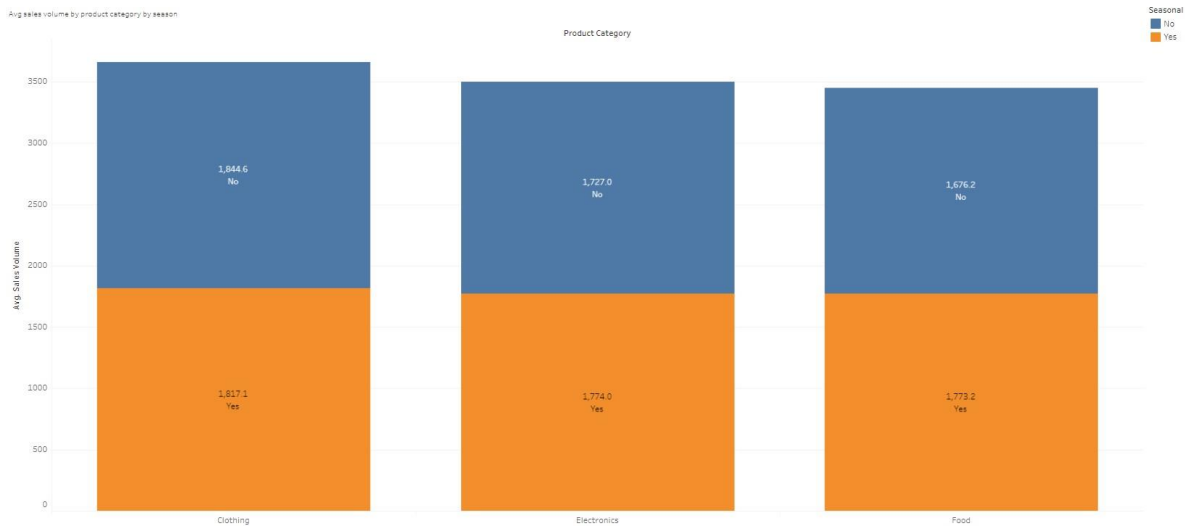
Plotting the percentage price premium over competitors against sales volume shows a very weak relationship (correlation of approximately -0.03). Most products carry a price premium between 4% and 18% versus competitors, and this premium does not show a clear linear effect on units sold across any product category.

Q5: Which consumer demographic segment drives the most sales?



Young adults generate the highest average sales volume (1,822 units), followed by College students (1,793) and Seniors (1,783). Families lag noticeably behind at 1,685 units — about 7% lower than Young adults — making Families the relatively weakest-performing demographic segment in this dataset.

Q6: Do seasonal products outperform non-seasonal products, and does this vary by category?



Across the full dataset, seasonal products average 1,789 units versus 1,749 for non-seasonal products, a modest uplift of about 2%. Breaking this down by category shows the effect is not uniform — some categories benefit more from seasonality than others, reinforcing that seasonality interacts with product type rather than acting as a universal driver.

Q7: Which combination of product category and shelf position performs best?



The strongest combination is Clothing at the Front of Store, averaging 1,924 units sold — noticeably above other category-position pairs. Clothing also performs well in Aisle placement (1,833 units), while Food at Front of Store is the weakest combination (1,673 units). This suggests merchandising strategy should be tailored by category rather than applying one shelfplacement rule across all products.

4. Key Takeaways

- The dataset is clean and analysis-ready, requiring no missing-value treatment or deduplication.
- No single factor (shelf position, promotion, foot traffic, or price premium) shows a strong standalone effect on sales volume — all differences are in the low single-digit percentage range.
- Demographic segment shows the most meaningful spread: Young adults outperform Families by roughly 7%.

- The clearest actionable signal is the category-by-position interaction: Clothing performs best at the Front of Store, while Food underperforms there, suggesting placement strategy should be category-specific rather than store-wide.
- Given the weak individual correlations, a multivariate model (e.g. regression or tree-based model) combining position, promotion, traffic, price, and demographics together would likely explain sales volume better than any single factor in isolation.